Wave Backend Design

What the app does

A user signs up or signs in, optionally sets up a voice (demo voice, recorded samples, or uploaded recordings), records a few seconds of silent speech on camera, and reviews the clip. The lipreading model returns about 5 possible phrases with confidence scores. The user picks one, can edit it, and the app speaks it in their voice. At the end the user marks the result as exactly right, right after editing, or not quite right, gives the voice 1 to 5 stars, and can leave a comment. A history page lists past phrases.

Services

Piece	Service	Job
Login	Amazon Cognito	Sign up, sign in, passwords.
Lipreading model	Auto-ASR lipreading model on the EC2 GPU server (L4)	Reads the silent video and returns the 5 possible phrases with scores.
Voice cloning	VoiceCraft model on the same GPU server	Clones the user's voice from their voice samples and speaks the chosen phrase.
API	FastAPI on the same server	Connects the app to the models and the database.
Database	RDS PostgreSQL	The 4 tables below.
Files	S3 (private bucket)	Videos, voice samples, spoken audio. The database only stores the file path.

Database: 4 tables

Four tables are enough. Each one matches a part of the app, and they connect through IDs: a voice and a translation both point to a user, and the 5 suggested phrases point to a translation. Feedback is stored in the translations table, since it is one answer per translation.

Table	Fields	Used by
users	user_id, username, created_at	Sign up and sign in. Passwords stay in Cognito, not here.
voices	voice_id, user_id, type (demo / recorded / uploaded), file_path, created_at	Voice setup and Voice samples pages.
translations	translation_id, user_id, voice_id, video_path, chosen_phrase, final_text, audio_path, model_version, time_taken_ms, accuracy (exact / edited / wrong), voice_rating, comment, created_at	Record, Review, Translating, Confirm phrase, Spoken in your voice, feedback, and History.
suggestions	suggestion_id, translation_id, rank (1 to 5), text, confidence	The 5 possible phrases on the Choose Phrase screen.

Endpoints

Endpoint	Purpose
POST /voices	Save a voice sample.
POST /translations	Upload video, run the lipreading model, return the 5 phrases.
POST /translations/{id}/choose	Save the chosen or edited phrase.
POST /translations/{id}/speak	Generate speech in the user's voice with VoiceCraft.
POST /translations/{id}/feedback	Save accuracy, stars, comment.
GET /history	The user's past translations.

Numbers we can report from these tables

Result	Where it comes from
Top-1 and top-5 accuracy	Whether the chosen phrase was rank 1, or any of ranks 1 to 5, in suggestions.
Exact / edited / wrong rates	translations.accuracy.
Speed	translations.time_taken_ms (median and 95th percentile).
Voice quality	translations.voice_rating, split by demo vs recorded voice.